Process Performance Qualification Protocol for the Manufacturing Process of LSN3832099/ASYM-140851 (Project Code CCL# LSN3832099/ASYM-140851) at Asymchem Life Science (Tianjin) Co., Ltd. (No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China)

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This document is reviewed and approved by Eli Lilly and Company. Refer to signature page for EDMS signatures and dates for Eli Lilly and Company approval.

Revision History

Effective Date	Protocol and version No.	Document Changes	Change Justification
	PVP-1400210301.01	The first version	N/A

Note：*The document shall only take effect after the training is completed, and the effective date will be manually filled in by designated personnel with signature and date once confirmed.*

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1. Introduction

N-(((5R,8S,11S,14S,17S,20R)-20-((S)-5-(tert-butoxy)-4-((tert-butoxycarbonyl)amino)-5-oxopentanamido)-8,14-bis((R)-1-(tert-butoxy)ethyl)-11-methyl-7,10,13,16,19-pentaoxo-17-(2-oxo-2-(tritylamino)ethyl)-1,3-dithia-6,9,12,15,18-pentaazacyclohenicosane-5-carbonyl)-L-alanyl)-O-(tert-butyl)-L-threonylglycine (the product is referred to as LSN4182928 in Lilly and ASYM-140021 in Asymchem) is an intermediate. The synthetic process is provided by Eli Lilly and Company (Lilly). Asymchem Life Science (Tianjin) Co., Ltd. (production address: No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China, referred to as Asymchem TJ4) is selected as the intended commercial supplier. LSN3832099/ASYM-140851 is a peptide on-resin in-process intermediate of the final product ASYM-140021, and the structure is shown in **Figure 1**.

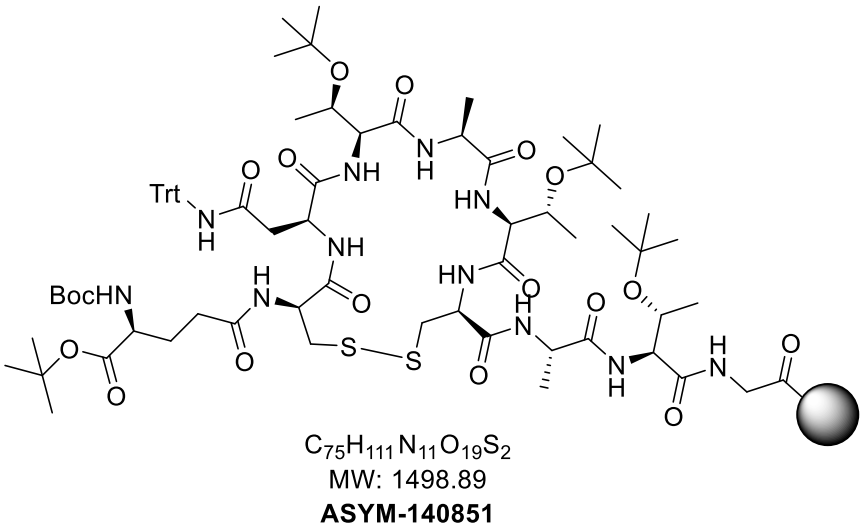


Figure 1: Structure of LSN3832099/ASYM-140851

The in-process intermediate LSN3832099/ASYM-140851 is prepared by solid-phase synthesis using 2-Chlorotrityl chloride (2-CTC) resin as the raw material. This synthetic route uses Fmoc-Gly-OH, Fmoc-L-Thr(tBu)-OH, Fmoc-L-Ala-OH·H_2O, Fmoc-L-Cys(Trt)-OH, Fmoc-L-Asn(Trt)-OH, Boc-L-Glu-OtBu as the proposed regulatory raw materials. The target compound is prepared by multiple cycles of solid-phase synthesis, washing, deswelling and drying. The synthetic route is described in **Figure 2** below.

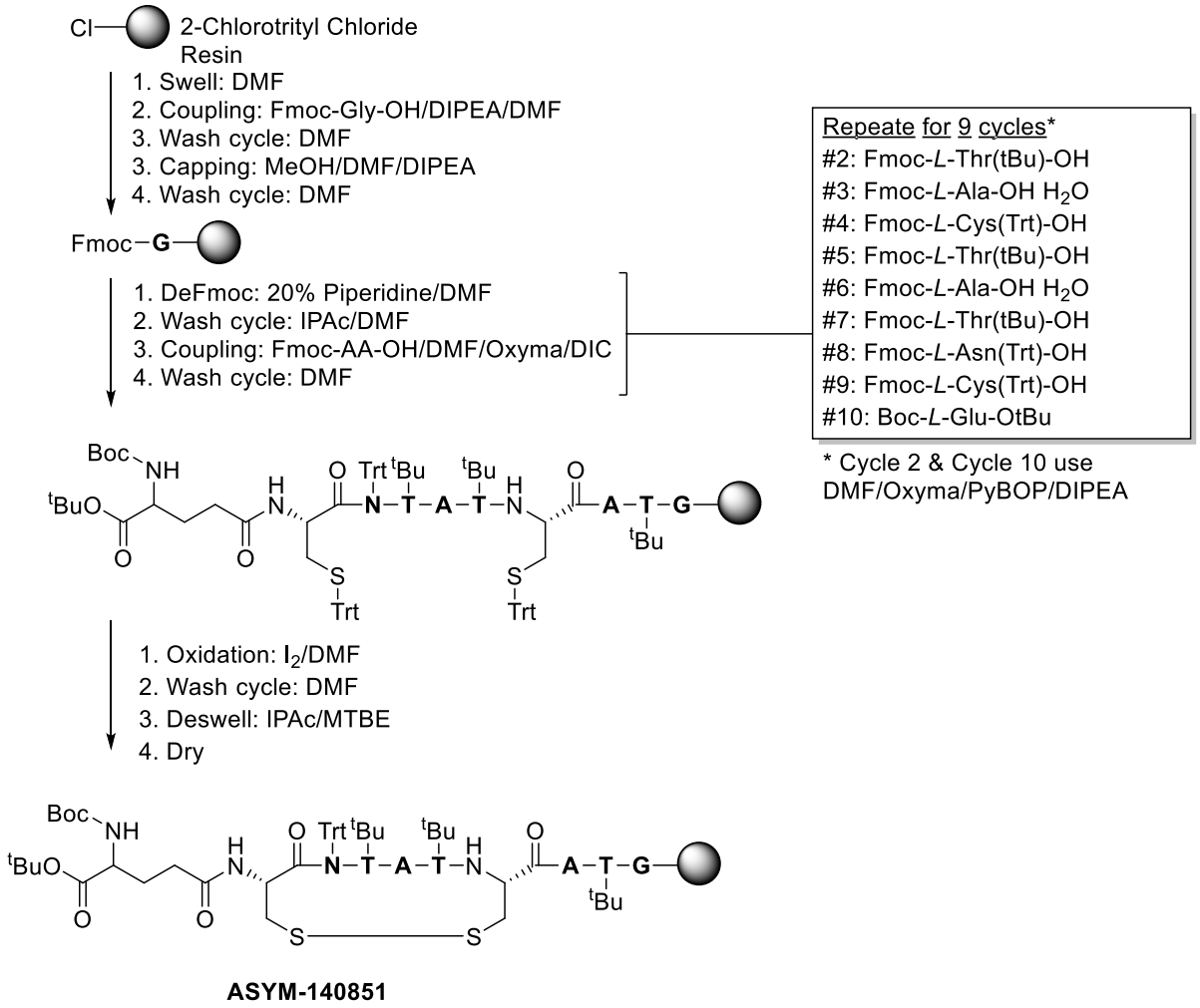


Figure 2: Synthetic route of ASYM-141851

The process was transferred from Lilly to Asymchem in August 2024, and the process was familiarized and further optimized for scaling up at Asymchem Life Science (Tianjin) Co., Ltd. (referred to as Asymchem TJ4) R&D laboratory. This Process Performance Qualification (PPQ) protocol has been developed as part of the overall process validation lifecycle strategy described in the applicable Validation Plan <Process Validation Master Plan for Manufacturing of LSN4182928 (ASYM-140021): VP-14002101.01 >. Process knowledge obtained during development, technology transfer, scale-up studies, risk assessments and process characterisation activities has been used to establish the manufacturing strategy and define the parameters requiring validation during PPQ. The process described in this protocol represents the intended commercial manufacturing process for LSN3832099/ASYM-140851 and PPQ execution is intended to demonstrate suitability of the process for routine commercial manufacture.

For current campaign CPo140021-03, the process performance qualification (PPQ) of LSN3832099/ASYM-140851 Step 1 is planned to be started in October 2026 under cGMP conditions at Asymchem TJ4. LSN3832099/ASYM-140851 is a peptide-on-resin in-process intermediate intended for further processing in downstream manufacturing operations. As the material is not an isolated intermediate with an approved release specification, process validation is primarily based on demonstrated control of the manufacturing process, successful achievement of in-process controls and suitability for downstream processing. For clarity and better understanding, the compound code and company code are listed in **Table 1** and **Table 2** respectively.

Table 1. Corresponding Asymchem and Customer compound codes

Step	RSM/Intermediate /Product	Chemical Name	Customer Material Code	Asymchem Code
1	RSM	Fmoc-Gly-OH	N/A	ASYM-138200
1	RSM	Fmoc-L-Thr(tBu)-OH	N/A	ASYM-138389
1	RSM	Fmoc-L-Ala-OH · H_2O	N/A	ASYM-138204
1	RSM	Fmoc-L-Cys(Trt)-OH	N/A	ASYM-138397
1	RSM	Fmoc-L-Asn(Trt)-OH	N/A	ASYM-138519
1	RSM	Boc-L-Glu-OtBu	N/A	ASYM-140413
1	In-process intermediate	N-(((4R,7S,10S,13S,16S,19R)-19-((S)-5-(tert-butoxy)-4-((tert-butoxycarbonyl)amino)-5-ox opentanamido)-7,13-bis((R)-1-(tert-butoxy)ethyl)-10-methyl-6,9,12,15,18-pentaoxo-16-(2-oxo-2-(tritylamino)ethyl)-1,2-dithia-5,8,11,14,17-pentaazacycloicosane-4-carbonyl)-L-alan yl)-O-(tert-butyl)-L-threonylglycine peptidyl resin	LSN3832099	ASYM-140851
2	Process intermediate	N-(((4R,7S,10S,13S,16S,19R)-19-((S)-5-(tert-butoxy)-4-((tert-butoxycarbonyl)amino)-5-ox opentanamido)-7,13-bis((R)-1-(tert-butoxy)ethyl)-10-methyl-6,9,12,15,18-pentaoxo-16-(2-oxo-2-(tritylamino)ethyl)-1,2-dithia-5,8,11,14,17-pentaazacycloicosane-4-carbonyl)-L-alan yl)-O-(tert-butyl)-L-threonylglycine	LSN4214882	ASYM-140852
3	Product	N-(((5R,8S,11S,14S,17S,20R)-20-((S)-5-(tert-butoxy)-4-((tert-butoxycarbonyl)amino)-5-oxopentanamido)-8,14-bis((R)-1-(tert-butoxy)ethyl)-11-methyl-7,10,13,16,19-pentaoxo-17-(2-oxo-2-(tritylamino)ethyl)-1,3-dithia-6,9,12,15,18-pentaazacyclohenicosane-5-carbonyl)-L-alanyl)-O-(tert-butyl)-L-threonylglycine	LSN4182928	ASYM-140021

Table 2. Abbreviation code of the related company name

Company Name	Company Address	Abbreviation
Eli Lilly and Company	Eli Lilly Kinsale Limited Dunderrow Kinsale Cork P17 NY71 Ireland	Lilly
Asymchem Life Science (Tianjin) Co., Ltd.	No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China	Asymchem TJ4
Chengdu ZY Biochemical Technology Co., Ltd.	No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China	ZY Biochem
Asymchem Laboratories (Jilin) Co., Ltd.	No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China	DH3

Company Name	Company Address	Abbreviation
Liaoning Asymchem Laboratories Co., Ltd.	Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China	FX2
Tianjin Nankai Hecheng S&T Co., Ltd.	No.969-6, Kaixuan Street, Dagang Petro chemical Industrial Area, Binhai New Area, Tianjin, P.R.China	Hecheng
Sunresin New Materials Co. Ltd.	No.135, jinye Road, Xi'an Hitech Industrial Development Zone	SUN RESIN

2. Objective and Scope

The objective of this protocol is to establish scientific and risk-based evidence which provides a high degree of assurance that the manufacture process of LSN3832099/ASYM-140851 will consistently produce LSN3832099/ASYM-140851 meets its IPC acceptance criteria and is suitable for forward processing.

The scope of this PPQ protocol is applicable to LSN3832099/ASYM-140851 manufacturing process using Fmoc-Gly-OH, Fmoc-L-Thr(tBu)-OH, Fmoc-L-Ala-OH · H_2O, Fmoc-L-Cys(Trt)-OH, Fmoc-L-Asn(Trt)-OH, Boc-L-Glu-OtBu as the proposed regulatory starting materials, with 2-CTC resin included as a critical raw material. Manufacturing will be performed at the defined commercial manufacturing scale of target 65.0 kg 2-CTC Resin per batch for the SPPS process in Building 12 at Asymchem TJ4 site. This protocol covers manufacture of LSN3832099/ASYM-140851 at commercial scale in the qualified equipment train identified within this document and includes evaluation of process performance against CPPs and IPCs. Cleaning validation, stability and supporting studies are performed alongside PPQ under separate protocols (Sections 16–18)

3. Validation Type and Strategy

A prospective Process Performance Qualification (PPQ) approach will be used to demonstrate that the commercial manufacturing process for LSN3832099/ASYM-140851, when operated within the established process parameters and control strategy, is capable of manufacturing material that reliably meets its predefined quality attributes, in-process requirements and meets its IPC acceptance criteria and is suitable for forward processing. The PPQ will be executed using the intended commercial manufacturing process, qualified facility, utilities and equipment, approved manufacturing instructions, qualified analytical methods, approved raw materials and appropriately trained personnel.

The validation strategy has been established using a science- and risk-based assessment of process knowledge, process complexity, sources of variability, and the relationship between material attributes/process parameters and product quality attributes. Based on the available development and manufacturing knowledge, process risk assessment and demonstrated process understanding, three consecutive PPQ batches manufactured at the intended commercial scale using the defined commercial equipment train are considered sufficient to provide evidence of process reproducibility and an initial state of control. All three PPQ batches will be manufactured in the same equipment train, in accordance with the approved Master Operation Record (Refer to Appendix 2) and this PPQ protocol (PVP-1400210301.01).

4. Manufacturing, release and stability testing site

The manufacturing, release testing and stability testing sites are indicated in **Table 3** below.

Table 3. Manufacturing, release and stability testing site

Step No.	Manufacturing site and address	Release testing site and address	Stability testing site and address
Step 1	Asymchem TJ4	Asymchem TJ4	Asymchem TJ4
	No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China	No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China	No.265, South Avenue, Binhai New Area TEDA, Tianjin, 300462, China

5. Functions and Responsibilities

CMMD

1) Responsible for overall planning and coordination of the PPQ;
2) Drafting, review and approval of PPQ document;
3) Monitoring the process implement, organization of PPQ activity.

Quality Assurance

1) Review and approval of PPQ documents;
2) Ensuring all validation activities are in accordance with Asymchem procedure and cGMP standards;
3) Organize evaluation of related change controls and deviations.

Production Department

1) Responsible for review and approval of PPQ documents;
2) Responsible for operation of production, cleaning of facility/equipment during PPQ;
3) IPC sampling.

Quality Control

1) Responsible for the review and approval of PPQ documents.
2) Ensure that specifications and validated analytical methods are available and relevant analytical instruments are qualified;
3) IPC testing, sampling and testing of product and intermediate, and equipment cleaning testing.

Lilly

1) Review and approve validation documents per the Quality Agreement.
2) Define the Validation Strategy and communicate APIEM process validation expectations.
3) Provide technical oversight for critical quality attributes and process parameters
4) Provide GMP oversight and ensure alignment with Lilly governing standards.

The personnel who are executing the protocol shall enter his/her name in training signature table. The training signature record shall be attached to report to indicate that the personnel are trained on the SOP in **Appendix 1** (current version).

6. Batch Plan

The planned batch information for PPQ is summarized in the table below.

Table 4. Batch information

Step	Vessel size	Type of batches	Number of batches	Batch Input*	Batch Output*	Yield range
1	3000 L	PPQ batch 1	CPo140021-03-01-02	2-Chlorotrityl chloride resin 65.00 kg target (64.87 – 65.13 kg)	LSN3832099/ASYM-140851 (157.69~292.80 kg)	80% - 130%
		PPQ batch 2	CPo140021-03-01-03			
		PPQ batch 3	CPo140021-03-01-04			

*Note: * Input varies within a ±0.2% range of the target input. The batch size (output) is calculated based on the input range and the expected yield range. The lowest input amount charged combined with the expected lowest yield shall give the output lower limitation, while the highest input amount charged combined with the expected highest yield shall give the output upper limitation.*

Process validation (PPQ) strategy: Figure 3 depicts the Eloralintide Fragment 3 PPQ batch genealogy. Three commercial-scale SPPS PPQ batches each generate two cleavage-cyclisation sub-batches. Under the "3+1" design, four sub-batches serve as Step 2 and Step 3 PPQ batches, while the second sub-batches of SPPS PPQ2 and SPPS PPQ3 are commercial batches. Different approved raw-material sources are used to evaluate resin and amino-acid impacts across the full Step 1-3 sequence. Both sub-batches from SPPS PPQ1 are designated as PPQ batches to demonstrate the reproducibility of Step 2 and Step 3 starting from a common peptide-on-resin input. Commercial sub-batches shall be manufactured against approved master batch records with equivalent sampling and testing requirements as PPQ batches prior to establishment of the ongoing process verification (OPV) plan. Batch-related governance items are defined within Asymchem PPQ protocol(s), which require Lilly API EM Quality approval prior to execution.

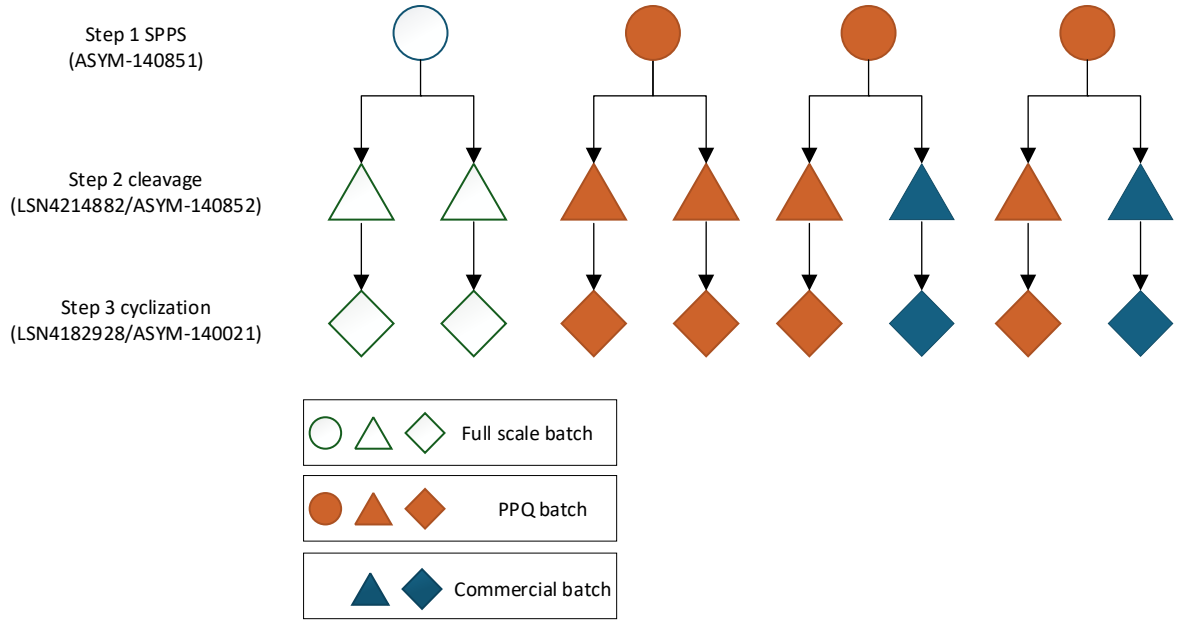


Figure.3: Batch tree of the PPQ batches manufacture

7. Qualified Process Equipment

All the major equipment to be used in this process performance qualification campaign have completed IQ, OQ and PQ (see Table 5). The equipment selected is fit for process requirement, referring to equipment suitability assessment <RA-140021030101.01> Equipment selection has been assessed through technology transfer and scale-up activities and is considered suitable for commercial manufacture of LSN3832099/ASYM-140851 (reference equipment suitability documentation: RA-140021030101.01).

All critical utilities supporting manufacture of PPQ batches have been qualified and are maintained in a qualified state through approved monitoring and maintenance programs.

Any modification to qualified equipment, automation systems, utilities or process control systems during PPQ execution shall be managed through the approved change control system and assessed for validation impact.

Table 5. Equipment Plan

Step No	Equipment	Size/L	MOC	Equipment ID	Location	Stirrer	Function in the process	Equipment Characteristics	MOC (All components in direct contact must be listed)	Specific Validation Protocol &Report No.
Step 1	Peptide Solid Phase Synthesizer	3000 L	Hastelloy	TJ4S-1213-Sy16	Building 12 module 13	4-bladed inclined paddle+ S-type impeller	Peptide synthesis, Fmoc deprotection, Wash resin, Drying	Stirring volume range of equipment (at rest): 36.69 L Minimum volume of the inner temperature probe (at rest): 26.71 L Control range of mixture temperature of the equipment capability: -12.8~52℃ Control range of jacket temperature of the equipment capability: -18~58℃ The pressure range of equipment for normal operation: -0.08~0.08 MPa The range of agitation speed of equipment capability: 5~40 rpm	**Hastelloy (C-276)** (inner surface, temperature probe, agitator, sieve plate) **316 SS** (pipeline, Mass flow meter**)** **PTFE** (Gasket, including manway gaskets and gaskets on all inlet valve manifolds) **316SS+PTFE** (Transfer pump, Waste discharge pump) **Outer coated FEP core silicone rubber** (sealing ring) **PTFE lined& Hastelloy** (valve) **Hastelloy (C-276) & FFKM** (mechanical seal) **Glass** (sight glass) **NRL** (gloves)	VR-TJ4-334.01-090.01

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Step No	Equipment	Size/L	MOC	Equipment ID	Location	Stirrer	Function in the process	Equipment Characteristics	MOC (All components in direct contact must be listed)	Specific Validation Protocol &Report No.
	Activation tank	1500 L	SS (S31603)	TJ4S-1213-Sy16-R01	Building 12 module 13	Double-layer four bladed inclined paddle type	Dissolve and amino acid activation	Stirring volume range of equipment (at rest): 57.4 L The temperature measurement of the equipment is from the bottom. Control range of temperature of the equipment capability: -6~54℃ Control range of temperature of the equipment capability: -14~55℃ The pressure range of equipment for normal operation: 0~0.08 MPa The range of agitation speed of equipment capability, 15~130 rpm	**316L** (inner surface, temperature probe, agitator, pipeline, valve) **PTFE (**gasket, including manway gaskets and gaskets on all inlet valve manifolds) **316SS+PTFE** (Transfer pump) **304L SS (mass flowmeter, pipeline, valve)** **Glass** (sight glass) **316L & FFKM** (mechanical seal)	VR-TJ4-334.01-090.01

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ASYMCHEM	Process Performance Qualification Protocol of LSN3832099/ASYM-140851	Project	CPo140021-03
		Protocol and version No.	PVP-1400210301.01

Step No	Equipment	Size/L	MOC	Equipment ID	Location	Stirrer	Function in the process	Equipment Characteristics	MOC (All components in direct contact must be listed)	Specific Validation Protocol &Report No.
	Agitation tank	500 L	Glass-lined	TJ4S-1213-AT09	Building 12 module 13	Impeller Agitator(3-blades impeller)	Iodine/DMF solution preparation	Stirring volume range of equipment (at rest): 39.50 L Minimum volume of the inner temperature probe (at rest): 55.30 L The range of agitation speed of equipment capability: 0-131.66 rpm Control range of temperature of the equipment capability: -12.2~53.9℃ The pressure range of equipment for normal operation: -0.090~0.08 MPa	**GL** (inner surface, agitator, sight glass) **GL&Tantalum (**temperature probe**)** **Tantalum** (Mass flowmeter) **Halar** (pump) **304 SS lined with thick-walled PTFE & Halar** (pipeline, backpressure valve**)** **PTFE&FFKM** (mechanical seal)	SVP -TJ4-170.01-064.01
	Agitation tank	500 L	Hastelloy	TJ4S-1200-AT01	Building 12 module 13	Propeller agitator (3-blades impeller)	DIC/DMF solution preparation	Stirring volume range of equipment (at rest): 19.95 L Minimum volume of the inner temperature probe (at rest): 68.75 L The range of agitation speed of equipment capability: 0-119.71 rpm The pressure range of equipment for normal operation: 0~0.08 MPa	**GL** (sight glass) **C-276** (inner surface, temperature probe, agitator) **316 SS** (Mass flowmeter, pump, valve) **PTFE** (sealing ring, gasket) **PTFE** (mechanical seal)	SVP -TJ4-170.01-067.01

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Step No	Equipment	Size/L	MOC	Equipment ID	Location	Stirrer	Function in the process	Equipment Characteristics	MOC (All components in direct contact must be listed)	Specific Validation Protocol &Report No.
	Agitation tank	3000L	SS (316L)	TJ4S-1213-SPS01-AT01	Building 12 module 13	Double-layer propeller	Deprotection solution preparation	Minimum volume of the stirrer can reach at rest: 308.7L; Minimum volume of the stirrer can reach at maximum speed: 289.2 L. Maximum working volume: 3000 L Minimum volume of the thermometer tube can reach at rest: 416.9 L; Minimum volume of the thermometer tube can reach at maximum speed: 356.8L.	**304 SS (**inner surface, temperature probe, agitator, pipeline, transfer pump, mass flowmeter, valve) **Glass** (sight glass) **PTFE** (gasket, including manway gaskets and gaskets on all inlet valve manifolds) **304SS & FFKM** (mechanical seal)	SVP-TJ4-392.01-002.01
	Agitation tank	3000L	SS (316L)	TJ4S-1213-SPS01-AT02	Building 12 module 13	Double-layer propeller	Deprotection solution preparation	Minimum volume of the stirrer can reach at rest: 309.3 L; Minimum volume of the stirrer can reach at maximum speed: 317.1 L. Maximum working volume: 3000 L Minimum volume of the thermometer tube can reach at rest: 376.1 L; Minimum volume of the thermometer tube can reach at maximum speed: 349.7 L	**304 SS (**inner surface, temperature probe, agitator, pipeline, transfer pump, mass flowmeter, valve) **Glass** (sight glass) **PTFE** (gasket, including manway gaskets and gaskets on all inlet valve manifolds) **304SS & PTFE & FFKM** (mechanical seal)	SVP -TJ4-392.01-002.01

Note: The gauges related to equipment are managed by electronic system CMS (Calibration Management System) to ensure the gauges are calibrated and valid before PPQ manufacturing.

8. Process Chemistry Overview

The process of Step 1 is based on the Lilly technical package and has been developed and finalized via risk assessment and the execution of designed experiments. The proven acceptable range (PAR) for all of the process parameters has been established <Refer to RPT- (Lilly document number: RPT-1193831.) >and form the basis of this process performance qualification. Only process parameters determined through process characterization and risk assessment to impact product quality or process performance, are considered within PPQ success criteria. Other operational parameters remain controlled through the approved MOR and are evaluated for potential impact should any deviation occur. The process flow chart is shown in **Figure 4**.

➢ Process Flow Chart

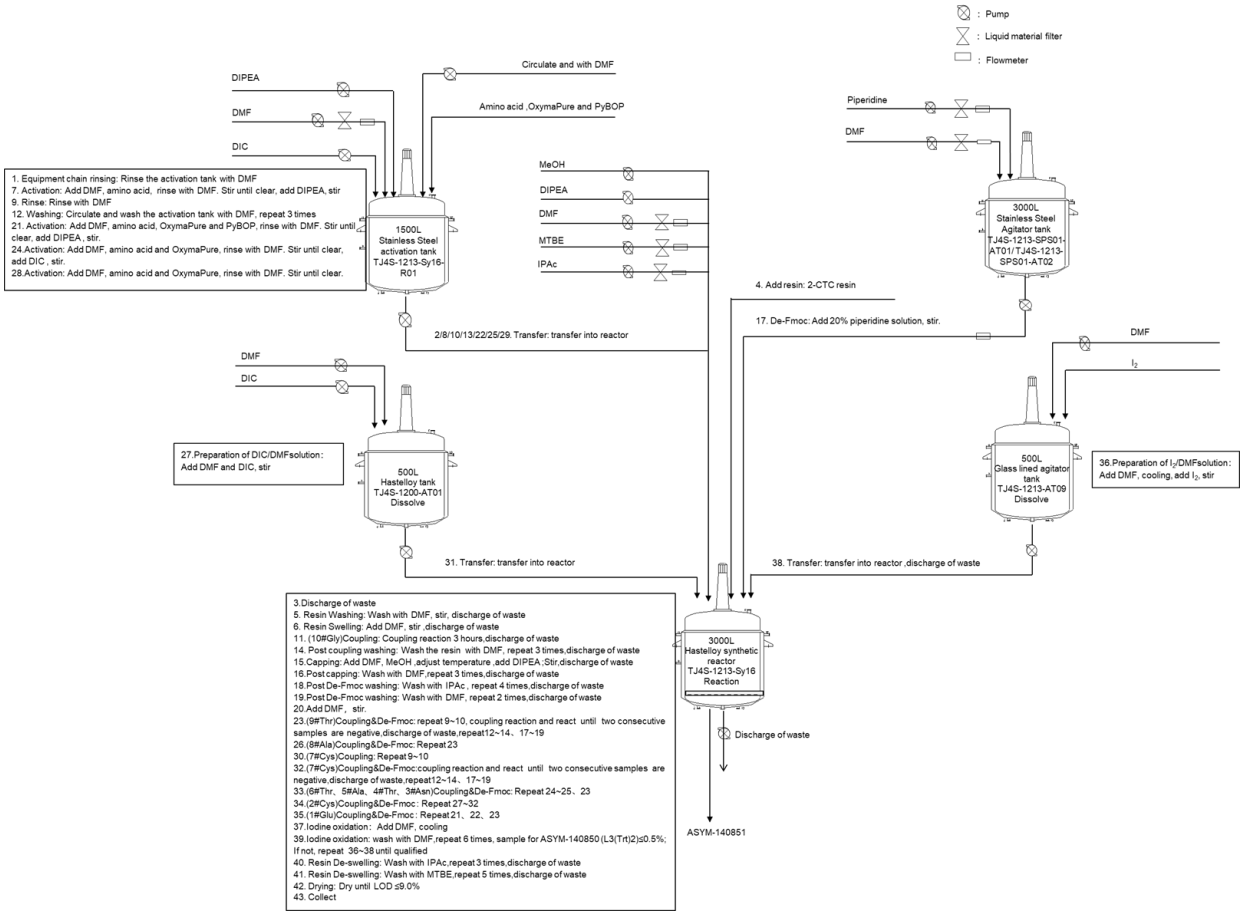


Figure 4. The Process Flow Chart of LSN3832099/ASYM-140851

➢ Process description

The process description is summarized in **Table 6** below.

Table 6. Process description for the process performance qualification

#	Unit operation name	Process Description
1.	Dispense & Solvent confirmation	**Materials involved in dispensing:** 2-Chlorotrityl Chloride Resin; Fmoc-Gly-OH; Fmoc-L-Thr(tBu)-OH; Fmoc-L-Ala-OH· H_2O; Fmoc-L-Cys(Trt)-OH; Fmoc-L-Asn(Trt)-OH; Boc-L-Glu-OtBu; Oxyma Pure; PyBOP; Elemental iodine.

#	Unit operation name	Process Description
		Solvent confirmation The 20.00%v/v Piperidine/DMF solution is prepared and the bottom valve of tank is opened. Confirmed that the DMF supply system from tank farm was supplied. Confirmed that the IPAC supply system from tank farm was supplied. Confirmed that the MTBE supply system from tank farm was prepared.
2.	Pre-treatment	Add 6.0 Vol DMF（5.69kg/kg） by spray balls into activation tank TJ4S-1213-Sy16-01, start the agitator, after stir for 10min±5min (target: 10min). Then transfer the solution into synthesis reactor TJ4S-1213-Sy16. Turn on the nitrogen and start the agitator, stop the stirring and turn off the nitrogen after stir for 10min±5min (target: 10min), sample the mixture for water content analysis and drain until there is no obvious dripping. If KF≤500ppm, move to next operation. If KF>500 ppm, repeat Module 3.0 until KF≤500 ppm. Note: In all the following operations, unless otherwise noted. The agitation speed of synthesis reactor is set as 20±10rpm (target: 20 rpm), the agitation speed of activation tank is set as 115±5 rpm (target: 115 rpm) and the nitrogen flow rate is set to 8.5±4m³/h (target: 8.5 m³/h).
3.	Resin swelling& Resin wash	Adjust the synthesis reactor jacket temperature TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). Add 65.00kg 2-chlorotrityl chloride resin (1.00eq) with powder feeding bags after dispensing. Rinse reactor wall with 8.0 Vol DMF (7.58kg/kg) by spray balls. Bubble with nitrogen for 5±2min (target: 5min) and start the agitator. Adjust to 25.0±5.0ºC. After stirring for time of 45min±15min (target: 45min), stop agitator, turn off the nitrogen. Drain until there is no obvious drip. Add 5.0Vol DMF (4.74kg/kg) by spray balls into synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
4.	Fmoc-Gly-OH activation	Add 6.0Vol DMF (5.69kg/kg) by spray balls into the activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-Gly-OH (2.25eq.) with powder feeding bags, rinse port with 1.0Vol DMF (0.95kg/kg) by spray balls, Adjust temperature of the mixture to 15.0±5.0ºC. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. at 15.0±5.0ºC (target: 15.0ºC) and later use. Add DIPEA (6.00eq) into activation tank TJ4S-1213-Sy16-R01 through metering pump at a reference rate of 40~80 kg/h at 15.0±5.0ºC (target: 15.0ºC). The addition rate depends on temperature control. After addition, stirring for 5± 5min (target: 5min).
5.	Fmoc-Gly-OH coupling	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 25.0±5.0ºC (target: 25.0ºC). Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0VolDMF (0.95kg/kg,) by spray balls into activation tank TJ4S-1213-Sy16 -R01to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture(DMF) from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust temperature to 25.0±5.0ºC in synthesis reactor TJ4S-1213-Sy16. The mixture is allowed to react at 25.0±5.0ºC (target: 25.0ºC, starts timing when the temperature reaches 20.0ºC) for 3h±15 min (target:3h). Drain until there is no reaction mixture drip. Move to next operation.
6.	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 8.0Vol DMF (7.58kg/kg by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered

#	Unit operation name	Process Description
		out of the synthesis reactor TJ4S-1213-Sy16, the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min(target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
7.	Resin Capping	Add 9.0Vol DMF (8.53kg/kg) by spray balls into synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), start the agitator. Add methanol (10.00eq) through peristaltic pump. Adjust the mixture to 20.0±3.0°C (target: 20.0°C). Add DIPEA (5.00eq)(reference rate of 40~80 kg/h) at 20.0±3.0°C. The addition rate depends on temperature control. After addition, adjust the mixture to 25.0±5.0°C (target: 25.0°C. Start timing when temperature reaches 20.0°C) The mixture is allowed to react at 25.0±5.0°C (target: 25.0°C) for 60min±30min (target: 60min), start timing when the temperature reaches 20.0°C. Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
8.	Wash resin	Add 8.0Vol DMF (7.58kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
9.	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0°C (target: 25.0°C). Add 8.0Vol 20% (V/V) piperidine/DMF (7.44kg/kg) solution into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0°C), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
10	Wash resin	Add 8.0Vol IPAc(6.98kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min). Then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
11	Re-swelling	Add 8.0Vol DMF(7.58kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
12	Fmoc-L-Thr(tBu)-OH activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0°C (target: 20.0°C). Add 4.0Vol DMF(3.79kg/kg,) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Thr(tBu)-OH(2.00eq) with powder feeding bags into activation tank TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust to 5.0±5.0°C. Add PyBOP (2.00eq) with powder feeding bags and Oxyma Pure (2.00eq) with powder feeding bags at 5.0±5.0°C (target: 5.0°C). Rinse reactor wall with 1.0Vol DMF(0.95kg/kg) by spray balls. Add DIPEA (3.00eq) throughmetering pump at a reference rate of 20~40 kg/h at 5.0±5.0°C (target: 5.0°C). The addition rate depends on temperature control. After addition. Stir for 20±5min (target: 20min).
13	Fmoc-L-Thr(tBu)-OH coupling	Add 3.0Vol DMF(2.84kg/kg) by spray balls into synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), start the agitator.

#	Unit operation name	Process Description
		Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0Vol DMF (0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). The mixture is allowed to react at 25.0±5.0 ºC (target: 25.0 ºC), start timing when the temperature reaches 20.0 ºC. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h), Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
14	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 8.0Vol DMF (7.58kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min(target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
15	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). Add 8.0Vol 20% (V/V) piperidine/DMF solution (7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0ºC), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
16	Wash resin	Add 8.0Vol IPAc (6.98kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min). Then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
17	Re-swelling	Add 8.0Vol DMF (7.58kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
18	Fmoc-L-Ala-OH·H_2O activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0ºC (target: 20.0ºC). Add 6.0Vol DMF (5.69kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Ala-OH· H_2O (2.00eq) with powder feeding bags after dispensing into activation tank TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust temperature to 20.0±5.0ºC (target: 20.0ºC). Add Oxyma Pure (2.00eq) with powder feeding bags after dispensing at 20.0±5.0ºC (target: 20.0ºC). Rinse reactor wall with 1.0Vol DMF (0.95kg/kg) by spray balls. Then add DIC(2.20eq) through metering pump at a reference rate of 40~80 kg/h. Add DIC at 20.0±5.0°C (target: 20.0ºC). Stir for 20min±5min (target: 20min) after addition.

#	Unit operation name	Process Description
19	Fmoc-L-Ala-OH·H_2O coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0VolDMF(0.95kg/kg,1.0Vol) by spray balls into activation tank TJ4S-1213-Sy16 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). The mixture is allowed to react at 25.0±5.0 ºC (target: 25.0 ºC), start timing when the temperature reaches 20.0 ºC. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h) Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
20	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 8.0Vol DMF (7.58kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min(target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
21	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). Add 8.0Vol 20% (V/V) piperidine/DMF solution (7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0ºC), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
22	Wash resin	Add 8.0Vol IPAc (6.98kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min). Then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
23	Re-swelling	Add 8.0Vol DMF (7.58kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
24	Prepare solution	Add 1.0Vol DMF (0.95kg/kg) into the Hastelloy agitator tank TJ4S-1200-AT01. Then add DIC(3.30eq) by metering pump. Stir for 15±5min (target: 15min) for later use. Reference agitation speed is 60±20 rpm (target: 60rpm).
25	Fmoc-L-Cys(Trt)-OH dissolution	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0ºC (target: 20.0ºC). Add 6.0Vol DMF (5.69kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Cys(Trt)-OH (2.00eq) with powder feeding bags after dispensing and Oxyma Pure(3.00eq) with powder feeding bags after dispensing into activation tank with powder feeding bags after dispensing. Rinse reactor wall with 1.0Vol DMF (0.95kg/kg) by spray balls. Maintain stirring for NLT 15min. Confirm no visible solids and later use. Adjust to 5.0±5.0°C. Stir for

#	Unit operation name	Process Description
		5±5min (target: 5min) at 5.0±5.0℃ (target: 5.0℃).
26	Fmoc-L-Cys(Trt)-OH coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0VolDMF(0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 20±5 ℃ (Target:20℃). Transfer the DIC/DMF solution (Prepared in Ins.24) into the mixture at 20±5 ℃ (Target:20℃).Then adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 20.0±5.0℃ (target: 20.0℃). The mixture is allowed to react at 20.0±5.0 ℃ (target: 20.0 ℃), start timing when the temperature reaches 15.0 ℃. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
27	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 8.0Vol DMF(7.58kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min(target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
28	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0℃ (target: 25.0℃). Add 8.0Vol 20% (V/V) piperidine/DMF solution (7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0℃. After stirring for 110~150min (target: 2h) at 25.0±5.0℃ (target: 25.0℃), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
29	Wash resin	Add 8.0Vol IPAc (6.98kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min). Then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
30	Re-swelling	Add 8.0Vol DMF (7.58kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
31	Fmoc-L-Thr(tBu)-OH activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0℃ (target: 20.0℃). Add 6.0Vol DMF(5.69kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Thr(tBu)-OH(2.00eq) with powder feeding bags after dispensing into activation tank TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust temperature to 20.0±5.0℃ (target: 20.0℃). Add Oxyma Pure (2.00eq) with powder feeding bags after dispensing at 20.0±5.0℃ (target: 20.0℃). Rinse reactor wall with 1.0Vol DMF (0.95kg/kg) by spray balls. Then add DIC through metering pump at a reference rate of 40~80 kg/h

#	Unit operation name	Process Description
		at 20.0±5.0℃ (target: 20.0℃). Stir for 20min±5min (target: 20min) after addition.
32	Fmoc-L-Thr(tBu)-OH coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0Vol DMF(0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0℃ (target: 25.0℃). The mixture is allowed to react at 25.0±5.0 ℃ (target: 25.0 ℃), start timing when the temperature reaches 20.0 ℃. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
33	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 8.0Vol DMF (7.58kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min (target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
34	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0℃ (target: 25.0℃). Add 8.0Vol 20% (V/V) piperidine/DMF (7.44kg/kg) solution into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0℃. After stirring for 110~150min (target: 2h) at 25.0±5.0℃ (target: 25.0℃), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
35	Wash resin	Add 8.0Vol IPAc(6.98kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min) then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
36	Re-swelling	Add 8.0Vol DMF (7.58kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
37	Fmoc-L-Ala-OH•H_2O activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0℃ (target: 20.0℃). Add 6.0Vol DMF(5.69kg/kg) by spray balls into activation tank. Start the agitator. Add Fmoc-L-Ala-OH · H_2O (2.00eq) with powder feeding bags after dispensing into TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust temperature to 20.0±5.0℃ (target: 20.0℃). Add Oxyma Pure (2.00eq) with powder feeding bags after dispensing at 20.0±5.0℃ (target: 20.0℃). Rinse reactor wall with 1.0Vol DMF (0.95kg/kg) by spray balls. Then add DIC (2.20eq) through metering pump at a reference rate of 40~80 kg/h. Add DIC at 20.0±5.0℃ (target: 20.0℃). Stir for 20min±5min (target:) after addition.

#	Unit operation name	Process Description
38	Fmoc-L-Ala-OH· H_2O coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0VolDMF(0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). The mixture is allowed to react at 25.0±5.0 ºC (target: 25.0 ºC), start timing when the temperature reaches 20.0 ºC. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
39	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 10.0Vol DMF(9.48kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min (target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
40	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). Add 8.0Vol 20% (V/V) piperidine/DMF solution(7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0ºC), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
41	Wash resin	Add 10.0Vol IPAc (8.73kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min) then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
42	Re-swelling	Add 10.0Vol DMF(9.48kg/kg by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
43	Fmoc-L-Thr(tBu)-OH activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0ºC (target: 20.0ºC). Add 6.0Vol DMF(5.69kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Thr(tBu)-OH (2.00eq)after with powder feeding bags dispensing into TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust temperature to 20.0±5.0ºC (target: 20.0ºC). Add Oxyma Pure (2.00eq) with powder feeding bags after dispensing at 20.0±5.0ºC (target: 20.0ºC). Rinse reactor wall with 1.0Vol DMF(0.95kg/kg) by spray balls. Then add DIC (2.20eq) through metering pump at a reference rate of 40~80 kg/hat 20.0±5.0°C (target: 20.0ºC). Stir for 20min±5min (target: 20min) after addition.

#	Unit operation name	Process Description
44	Fmoc-L-Thr(tBu)-OH coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0VolDMF(0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). The mixture is allowed to react at 25.0±5.0 ºC (target: 25.0 ºC), start timing when the temperature reaches 20.0 ºC. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
45	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 10.0Vol DMF (9.48kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid to in activation tank is transferred the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min (target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
46	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). Add 8.0Vol 20% (V/V) piperidine/DMF solution (7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0ºC), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
47	Wash resin	Add 10.0Vol IPAc(8.73kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min) then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
48	Re-swelling	Add 10.0Vol DMF (9.48kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
49	Fmoc-L-Asn(Trt)-OH activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0ºC (target: 20.0ºC). Add 6.0Vol DMF (5.69kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Asn(Trt)-OH(2.00eq) with powder feeding bags after dispensing into TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust temperature to 20.0±5.0ºC (target: 20.0ºC). Add Oxyma Pure (2.00eq) with powder feeding bags after dispensing at 20.0±5.0ºC (target: 20.0ºC). Rinse reactor wall with DMF (0.95kg/kg) b by spray balls. Then add DIC through metering pump at a reference rate of 40~80 kg/h. Add DIC (2.20eq) at 20.0±5.0°C (target: 20.0ºC). Stir for 20min±5min (target: 20min) after addition.

#	Unit operation name	Process Description
50	Fmoc-L-Asn(Trt)-OH coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0VolDMF（0.95kg/kg） by spray balls into activation tank TJ4S-1213-Sy16-R01to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). The mixture is allowed to react at 25.0±5.0 ºC (target: 25.0 ºC), starts timing when the temperature reaches 20.0 ºC. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
51	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 10.0Vol DMF (9.48kg/kg by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min(target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
52	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0ºC (target: 25.0ºC). Add 8.0Vol 20% (V/V) piperidine/DMF solution(7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0ºC), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
53	Wash resin	Add 10.0Vol IPAc(8.73kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min) then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
54	Re-swelling	Add 10.0Vol DMF (9.48kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
55	Prepare solution	Add 1.0Vol DMF(0.95kg/kg) into the Hastelloy agitator tank TJ4S-1200-AT01. Then add DIC(3.30eq) by metering pump. Stir for 15±5min (target: 15min) for later use. Reference agitation speed is 60±20 rpm (target: 60rpm)
56	Fmoc-L-Cys(Trt)-OH dissolution	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0ºC (target: 20.0ºC). Add 6.0Vol DMF(5.69kg/kg) by spray balls into activation tank TJ4S-1213-Sy16-R01. Start the agitator. Add Fmoc-L-Cys(Trt)-OH (2.00eq) with powder feeding bags after dispensing and Oxyma Pure (3.00eq) with powder feeding bags after dispensing into activation tank with powder feeding bags after dispensing. Rinse reactor wall with 1.0Vol DMF(0.95kg/kg) by spray balls. Maintain Stirring

#	Unit operation name	Process Description
		for NLT 15min. Confirm no visible solids and later use. Adjust to 5.0±5.0°C. Stir for 5±5min (target: 5min) at 5.0±5.0°C (target: 5.0°C).
57	Fmoc-L-Cys(Trt)-OH coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0Vol DMF(0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16.Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 20±5 °C (Target:20). Transfer the DIC/DMF solution (Prepared in Ins.55) into the mixture at 20±5 °C (Target:20). Then adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 20±5°C (target: 20°C). The mixture is allowed to react at 20.0±5.0 °C (target: 20.0 °C), start timing when the temperature reaches 15.0 °C. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
58	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 10.0Vol DMF (9.48kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min(target:5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
59	De-Fmoc	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0°C (target: 25.0°C). Add 8.0Vol 20% (V/V) piperidine/DMF solution (7.44kg/kg) into the synthesis reactor. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer, adjust to 25.0±5.0°C. After stirring for 110~150min (target: 2h) at 25.0±5.0°C (target: 25.0°C), stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
60	Wash resin	Add 10.0Vol IPAc(8.73kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min) then start the stirrer. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 4 times.
61	Re-swelling	Add 10.0Vol DMF(9.48kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
62	Boc-L-Glu-OtBu activation	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to 20.0±5.0°C (target: 20.0°C). Add 6.0Vol DMF (5.69kg/kg) by spray balls into activation tank. Start the agitator. Add Boc-L-Glu-OtBu (2.00eq) with powder feeding bags after dispensing into activation tank TJ4S-1213-Sy16-R01. Maintain Stirring for NLT 15min. Confirm no visible solids and later use. Adjust to 5.0±5.0°C. Add PyBOP (2.00eq) with powder feeding bags after dispensing and Oxyma Pure (2.00eq) with powder feeding bags after dispensing at 5.0±5.0°C (target: 5.0°C). Rinse reactor wall with 1.0VolDMF (0.95kg/kg) by spray balls. Add DIPEA (3.00eq) through metering pump at

#	Unit operation name	Process Description
		a reference rate of 20~40 kg/h at 5.0±5.0℃ (target: 5.0℃).The addition rate depends on temperature control. After addition. Adjust the temperature to 20.0±5.0℃ (target: 20.0℃), and Stir for 20±5min (target: 20min).
63	Boc-L-Glu-OtBu coupling	Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Add 1.0Vol DMF(0.95kg/kg) by spray balls into activation tank TJ4S-1213-Sy16 to rinse the wall. Bubble with nitrogen for 5±2min (target: 5min), then start the agitator. Transfer the mixture from the activation tank TJ4S-1213-Sy16-R01 to the synthesis reactor TJ4S-1213-Sy16. Adjust the mixture in synthesis reactor TJ4S-1213-Sy16 to 25.0±5.0℃ (target: 25.0℃). The mixture is allowed to react at 25.0±5.0 ℃ (target: 25.0 ℃), start timing when the temperature reaches 20.0 ℃. After 1h±15min (target: 1h), sample the mixture every 1h±15min for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative (Negative: the resin is yellow to orange, react completely; Positive: the resin is blue to black, react incompletely) and the reaction has stirred for 4h±30min (target: 4h). Stop the stirrer and turn off the nitrogen. Drain until there is no drip.
64	Wash activation tank and wash resin	Rinse the activation tank TJ4S-1213-Sy16-R01 with 10.0Vol DMF (9.48kg/kg) by spray balls. Circulate and clean the activating liquid transfer pipeline, solid charging port cleaning pipeline and liquid charging port cleaning pipeline, clean every pipeline for 2±1min. After the liquid is filtered out of the synthesis reactor TJ4S-1213-Sy16 and transfer the liquid in activation tank is transferred to the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen, after 5±2min (target: 5min), start the agitator. After stirring for 10±2min (target: 10min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
65	Prepare I_2/DMF solution	Add 4.0VolDMF (3.79kg/kg) into glass lined agitator tank TJ4S-1213-AT09. Adjust to -5.0±5.0℃ (target: -5.0℃). Add Iodine (5.00eq) with powder feeding bags after dispensing in portions, with an interval of 5–10 minutes between each portion. Stir for 15±5min (target: 15min) for later use. Reference agitation speed is 60±20 rpm (target: 60rpm).
66	Oxidation	Add 12.0Vol DMF (11.37kg/kg) into synthesis reactor TJ4S-1213-Sy16 by spray balls. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. Adjust to -5.0±5.0℃ (target: -5.0℃). Transfer the I_2 /DMF solution into synthesis reactor TJ4S-1213-Sy16 at -5.0±5.0℃ (target: -5.0℃) through diaphragm metering pump metering within 30-80min (target: 50min). After addition, stop stirring and turn off the nitrogen. Drain until there is no obvious drip (0~25min (target: 0min)).
67	Wash resin	Adjust the jacket temperature of activation tank TJ4S-1213-Sy16-R01 to –5.0~0 ℃ (target: -3 ℃). Add 10.0VolDMF (9.48kg/kg, 10.0Vol) by spray balls into activation tank TJ4S-1213-Sy16-R01. Adjust the mixture to- –5.0~0℃ (target: -3℃). Later for use Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to –5.0~0℃ (target: -3℃). Transfer the mixture in activation tank TJ4S-1213-Sy16-R01 into synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 2 times.
68	Wash resin	Adjust the jacket temperature of synthesis reactor TJ4S-1213-Sy16 to 20.0±5.0℃ (target: 20.0℃). Add 10.0Vol DMF (9.48kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min), then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this

#	Unit operation name	Process Description
		operation 4 times. Take 5~10g resins and submit for analysis testing. If ASYM-140850 ≤0.5%, move to next operation. If ASYM-140850 (L3 $(Trt)_2$)>0.5%, repeat Ins.65~68.
69	Wash resin	Add 10.0Vol IPAc(8.73kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min). Then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 3 times.
70	Deswelling	Add 10.0Vol MTBE(7.41kg/kg) by spray balls into the synthesis reactor TJ4S-1213-Sy16. Bubble with nitrogen for 5±2min (target: 5min). then start the stirrer. After stirring for 15±2min (target: 15min), stop stirring, turn off the nitrogen. Drain until there is no obvious drip. Perform this operation 5 times.
71	Drying	Sweep with nitrogen above, nitrogen flow is 10m³/h±4m³/h (target: 10 m³/h), and vacuum underneath at T_{jacke}≤28.0°C(target： 25.0°C). After 12h, sample the solid for LOD analysis every 12±4h until LOD ≤ 9.0%. Note 1: After drying 6h, begin to stir the solid for 2~5 min at a speed of 10±5 rpm (target:10rpm) every 2±1h. Note 2: After sample the solid for LOD analysis. And the sample send to QC lab for testing.
72	Collect	Put Boc-L3-SS peptidyl resin (LSN3832099/ASYM-140851) into double bagged with LDPE (low-densitypolyethylene) bags. The inner pharmaceutical-grade LDPE powder charging bag is sealed with clips, and the outer pharmaceutical-grade LDPE bag is sealed with cable tie separately. Purge the inside of the package with nitrogen, seal for temporary storage. HPDE drum with cover for storage. Then placed in seal and avoid light and cold storage where the temperature is 2~8ºC

9. Critical Process Parameters and Critical Quality Attributes(CPPs and CQAs)

LSN3832099/ASYM-140851 (peptide on-resin) is an in-process intermediate without a release specification. It is not isolated as a bulk material and is forward processed directly into the Step 2 soft cleavage. Therefore, no critical quality attributes (CQAs) are defined for LSN3832099/ASYM-140851.

The quality of the Step 1 output is assured by (1) operating the critical process parameters (CPPs) within their proven acceptable ranges (PARs), and (2) in-process controls (IPCs) that confirm the peptide on-resin is suitable for forward processing to produce Step 2 intermediate ASYM-140852 and Step 3 product ASYM-140021 (LSN4182928) meeting their specifications.

The CPPs were identified based on laboratory experiments, development experience, previous campaign knowledge. The CPPs were classified according to their potential impact on the quality attributes of the downstream intermediate and Fragment L3. The CPPs, with their PARs, normal operating ranges and targets, are listed in **Table 7** No critical in-process controls (CIPCs) are established for Step 1.

The IPCs defined in **Table 9** shall be performed for each PPQ batch. Processing shall not proceed to the next operation until the IPC acceptance criterion is met (refer to **Table 9** for actions in the event of failure). IPC results will be summarized in Appendix 3.

- **Operating Ranges for Critical Process Parameters**

Table 7: Eloralintide fragment L3 Critical Process Parameters (CPPs)

Step	Parameter	Type	Target	PAR	NOR(MOR)	Unit
Step 1 SPPS	Gly10 Coupling temperature	CPP	25.0	**17.0 – 33.0**	20.0~30.0	℃
	Post Gly10 capping DIPEA addition temperature	CPP	20.0	**15.0 – 25.0**	17.0~23.0	℃
	Piperidine in DMF quantity (Thr9)	CPP	7.44	**6.58** – 8.39	7.22~7.66	kg/kg
	Piperidine in DMF quantity (Asn3)	CPP	7.44	**6.77** – 8.40	7.22~7.66	kg/kg
	DeFmoc temperature (Asn3)	CPP	25.0	**16.3** – 37.0	20.0~30.0℃	℃
	Total DeFmoc time (Thr6)	CPP	120	**62** – 180	110~150	min
	Total DeFmoc time (Asn3)	CPP	120	**77** – 180	110~150	min
	AA quantity (Cys7, Cys2)	CPP	2.00	**1.85 – 2.13**	1.94~2.06	eq
	DMF activation total quantity (Thr9)	CPP	4.74	4.14 – **5.27**	4.60~4.88	kg/kg
	DMF activation total quantity (Ala8, Thr6 to Asn3)	CPP	6.63	5.83 – **7.38**	6.43~6.83	kg/kg
	DMF activation total quantity (Cys7, Cys2)	CPP	6.63	5.81 – **7.38**	6.43~6.83	kg/kg

Step	Parameter	Type	Target	PAR	NOR(MOR)	Unit
	DMF coupling total quantity (Thr9)	CPP	8.53	7.45 – **9.49**	8.27~8.79	kg/kg
	DMF coupling total quantity (Ala8, Thr6 to Asn3)	CPP	7.58	6.68 – **8.43**	7.35~7.81	kg/kg
	DMF coupling total quantity (Cys7, Cys2)	CPP	8.53	**7.71– 9.26**	8.27~8.79	kg/kg
	Coupling temperature (Thr9, ᵞGlu1)	CPP	25.0	**17.0 – 33.0**	20.0~30.0	℃
	Coupling temperature (Ala8, Thr6 to Asn3)	CPP	25.0	**17.1 – 33.0**	20.0~30.0	℃
	Coupling temperature (Cys7, Cys2)	CPP	20.0	**12.0 – 28.0**	15.0~25.0	℃
	Oxidation temperature	CPP	-5	**-15 – 5**	-10.0~0.0	℃
	Iodine solution transfer time	CPP	50	**20 – 100**	30~80	min
	Oxidation Reaction time	CPP	0	0 – **30**	0~25	min
	Oxidation Exposure time	CPP	50	**20** – 130	30~105	min
	1st Post oxidation wash temperature	CPP	-3	**-7.0 – 5.0**	-5.0~0.0	℃

Asymchem confidential information

10. Materials and Specifications

➢ Origin of Materials

No starting materials, raw materials, reagents or solvents used in the manufacture of LSN3832099/ASYM-140851 as well as any of the intermediates are derived from human or animal sources. Asymchem will provide a BSE/TSE statement for each lot manufactured. All starting materials and critical raw materials used in PPQ batches shall be procured from approved suppliers and released against approved specifications prior to use.

➢ Raw Materials, Reagents and Solvents

All raw materials, solvents and reagents are sourced and controlled by Asymchem against current specifications. For details of the specifications, see: Material specifications for dedicated project RMS-TJ414002103 (current version).

➢ Regulatory Starting Material / Critical Raw Material

Fmoc-Gly-OH, Fmoc-L-Thr(tBu)-OH, Fmoc-L-Ala-OH· H2O, Fmoc-L-Cys(Trt)-OH, Fmoc-L-Asn(Trt)-OH and Boc-L-Glu-OtBu are identified as regulatory starting materials. 2-Chlorotrityl Chloride (2-CTC) resin is identified as critical raw material.

The vendors selected for the process performance qualification campaign are listed in **Table 8**.

Table 8. Suppliers of the starting material and critical material

No.	Amino acid	Control No.	CAS	Specification No.	Supplier	Manufacturer
1	2-Chlorotrityl chloride resin	YL003422	42074-68-0	RMS-TJ414002103 (current version)	Tianjin Nankai Hecheng S&T Co., Ltd.(Address: No.969-6, Kaixuan Street, Dagang Petro chemical Industrial Area, Binhai New Area, Tianjin, P.R.China)	Tianjin Nankai Hecheng S&T Co., Ltd. (Address: No.969-6, Kaixuan Street, Dagang Petro chemical Industrial Area, Binhai New Area, Tianjin, P.R.China)
					Sunresin New Materials Co. Ltd.(Address: No.135, jinye Road, Xi'an Hitech Industrial Development Zone)	Sunresin New Materials Co. Ltd. GaoLing(Address: No.999, Weiyang 9th Road, Gaoling District, Xi'an City Shaanxi)
2	Fmoc-Gly-OH	YL004846	29022-11-5	SP-138200-02 (current version)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)
					Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)

No.	Amino acid	Control No.	CAS	Specification No.	Supplier	Manufacturer
					Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)	Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)
3	Fmoc-L-Thr(tBu)-OH	YL003563	71989-35-0	SP-138389-02 (current version)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)
					Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)
					Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)	Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)
4	Fmoc-L-Ala-OH·H_2O	YL006160	207291-76-7	SP-138204-02 (current version)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)
					Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)
					Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)	Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)
5	Fmoc-L-Cys(Trt)-OH	YL003565	103213-32-7	SP-138397-03	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu

No.	Amino acid	Control No.	CAS	Specification No.	Supplier	Manufacturer
				(current version)	Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)	Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)
					Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)
					Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)	Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)
6	Fmoc-L-Asn(Trt)-OH	YL004843	132388-59-1	SP-138519-02 (current version)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)	Liaoning Asymchem Laboratories Co., Ltd (FX2).(Address: Yimatu Village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129,China)
					Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)
					Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)	Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, P.R.China)
7	Boc-L-Glu-OtBu	YL008618	24277-39-2	SP-140413-02 (current version)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)	Asymchem Laboratories (Jilin) Co., Ltd. (DH3) (Address: No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin 133700, China)
					Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County,	Chengdu ZY Biochemical Technology Co., Ltd.(Address: No.170 Qingpu Road, Shou'an Subdistrict, Pujiang County,

No.	Amino acid	Control No.	CAS	Specification No.	Supplier	Manufacturer
					Chengdu City, Sichuan Province, P.R.China)	Chengdu City, Sichuan Province, P.R.China)

11. Analytical Methods and Testing

- ***Analytical Methods Validation***

The analytical methods were developed and validated at Asymchem TJ4. All analytical methods for RSM and critical raw materials are validated before material release, refer to Change-TJ420260080.01; All IPC methods are validated before PV start, refer to AM-YL000336-004-VP-01, AM-140021-011-VP-01, AM-140021-043-VP-01, Asymchem shall follow the validated methods in the process performance qualification production

- ***Sample plan for IPC***

The IPC plan is developed by Asymchem and shall be used in the process performance qualification production.

The IPC tests are defined below：

IPC: In Process Control - Operations shall not proceed to the next step until results meeting the acceptance criteria are obtained; IPC targets are specified in **Table 9.**

Table 9. The testing plan of IPCs

No.	Process Step	Test	Analysis method	Acceptance Criteria	Action in the event of failure
1	Pre-treatment	KF	AM-YL000336-004	NMT 500 ppm	Recharge DMF for rinsing again until meeting the criteria
2	Amino acid coupling Completion (AA2~AA10)	Amino acid coupling Completion (AA2~AA10)	AM-140021-043	Negative	Continue stirring and sample the mixture for Bromophenol-blue colorimetric test analysis until two consecutive samples test as negative .
3	ASYM-140850 in post oxidation washing	ASYM-140850	AM-140021-011	≤0.5%	Repeat the iodine oxidation and wash procedures until meeting the criteria
4	Drying	Drying endpoint LOD	AM-140021-007	≤9.0%	Continue drying and sample the solid for LOD analysis until LOD≤9.0%

Note: In method AM-140021-007, vacuum drying oven is used for the LOD testing, it is simple technique that require no validation or verification.

12. Batch Production Record (OR) Review

Procedure: The CMMD, Production Department and Quality Assurance will review and approve each entry in all sections of the executed batch production record for the process performance qualification batches. Ensure that the information is correct, complete, and all appropriate approvals are performed.

13. Sample Retaining

LSN3832099/ ASYM-140851 is the peptide-on-resin in-process intermediate produced by the SPPS process in the PPQ batches and has no release specification. It shall be retained when a peptidyl resin stability study is required and sample retention is necessary. Samples will be taken for the purposes listed in **Table 10**:

Table 10. Sample Retains of LSN3832099/ASYM-140851

Sample	Sample quantity	Storage location	Storage condition.
Stability test *	~ 125 g	QC (Asymchem TJ4)	Refer to Stability Study Protocol PSP-140021-2026-002.01

Note: *In-process intermediate LSN3832099/ASYM-140851 from only one PPQ batch will be put into the stability chamber for stability.

14. Packaging and Storage

The material shall be packaged according to the following description after release Packaging specifications are summarized in **Table 11**.

Inner Packaging:

Double bagged with LDPE (low-density polyethylene) bags. The inner pharmaceutical-grade LDPE powder charging bag is sealed with clips, and the outer pharmaceutical-grade LDPE bag is sealed with cable tie separately. Purge the inside of the package with nitrogen. Place desiccant between the two layers (When Net weight≤1kg, NLT desiccant≥10% of Net weight ; Net weight＞1kg, desiccant =1% of Net weight+100g), seal for temporary storage.

Outer packaging:

HPDE drum with cover for storage.

Storage: seal and avoid light. Store at 2~8 °C.

Table11. Packaging Specifications

Packaging	Specification	Seal Type
Primary package	Size: 30 L or 50 L Pharmaceutical-Grade Powder feeding bag	Each bag is sealed with a sealing clip
Secondary package	Size: 1100 L × 700 W × 0.09 T (mm) Pharmaceutical-Grade LDPE bag	Each bag is sealed with a cable tie

Packaging	Specification	Seal Type
Outer packaging	Size: 50 L HPDE drum with cover for storage	Each drum is sealed with clips.
Desiccants	Size: W:78±4 mm L:150±7 mm Weight: 100 g Material: Silica Gel	desiccant placed between primary and secondary package

15. Stability Test

LSN3832099/ASYM-140851 peptide-on-resin is an in-process product without release specification that is further processed LSN3832099/ASYM-140851 in the downstream soft cleavage step 2. To support the stable hold time of peptide on-resin under drying conditions and its extended storage, a stability study will be established on PPQ batches. The stability monitoring plan is detailed in the LSN3832099/ASYM-140851 stability protocol<Refer to PSP-140021-2026-002.01>.

16. Homogeneity study

LSN3832099/ASYM-140851 As LSN3832099/ASYM-140851 is not an isolated bulk material and remains bound to the resin support, a conventional bulk homogeneity study is not considered applicable. Process consistency and uniformity are assured through control of the solid-phase synthesis process, including validated mixing capability of the reactor system, defined process parameters, in-process controls, and process characterisation knowledge. Therefore, a separate homogeneity study is not required for LSN3832099/ASYM-140851.

17. Cleaning Validation

Cleaning validation of the Step 1 equipment train is performed concurrently with the PPQ campaign under CVP-1400210301.01, in accordance with Asymchem's Cleaning Validation procedure (DS-GMP-043). The cleaning procedures, residue limits, and sampling and analytical methods are approved before PPQ starts. After each cleaning run, QA releases the equipment for reuse once results meet the protocol acceptance criteria. Batches manufactured after a cleaning validation run are released for further processing only after QA approves the interim cleaning validation report for that run. Cleaning validation results are reported in the cleaning validation report

18. Changes and Deviations

During the process performance qualification execution, any change or deviation occurring during the execution shall be handled based on the agreement<Quality argeement: QAA03550.V01> made between Asymchem and Lilly. Each event shall be assessed for impact on process validation conclusions and summarised in the validation report. After the campaign, these changes and deviations shall be reviewed in the final PPQ report.

19. Acceptance Criteria

LSN3832099/ASYM-140851 (peptide on-resin) is an in-process intermediate without a release specification, and no CQAs are defined for Step 1. In accordance with the stepwise process validation strategy defined in the Validation Plan (VP-14002101.01), Step 1 PPQ will be considered successful when

all of the following criteria are met for each of the three SPPS PPQ batches (CPo140021-03-01-02, CPo140021-03-01-03 and CPo140021-03-01-04):

The batches are manufactured according to the approved Master Operation Record (MOR-TJ414002101.04) and this protocol, using the qualified equipment train in **Table 5** and starting materials and critical raw materials from approved suppliers, released against approved specifications.

- All CPPs are maintained within the PARs defined in **Table 7**.
- All IPCs meet the acceptance criteria in **Table 9**.
- All deviations and changes during PPQ execution are handled under the Quality Agreement (QAA03550.V01). Each one is investigated and assessed as having no impact on product quality or on the validation conclusion.
- Any failure to meet a criterion shall be investigated under the deviation procedure. If the root cause is shown to be unrelated to the process, the process validation may continue by adding an additional batch once the issue has been remediated. The affected batch(es) are excluded from the PV data set but are discussed in the PV summary.

20. Summary Report

Upon completion of three SPPS PPQ batches, a Step 1 PPQ summary report shall be prepared. In line with the stepwise process validation strategy (VP-14002101.01), the report does not depend on Step 2 or Step 3 PPQ being completed. It shall be reviewed and approved according to the Quality Agreement (QAA03550.V01) and shall include:

Validation documentation summary: this protocol, the Master Operation Record, the executed batch production records, and referenced documents with their version numbers, including the cleaning validation protocol and report numbers.

Batch genealogy and materials:

The lots and suppliers of starting materials and 2-CTC resin used in each PPQ batch;

The substitution values of each resin lot and the total resin moles used as the charge basis;

The Step 2 and Step 3 sub-batches each PPQ batch is designated for.

Batch input, output and yield for each batch, compared with the ranges **in Table 4**.

CPP results for each batch, compared with the PARs (**Appendix 4**).

IPC results for each batch (**Appendix 3**), including any repeat operations allowed by the Master Operation Record.

Deviations, changes, OOS and ATR events, with their impact on the validation conclusion.

Conclusion on whether the process was executed reproducibly and in a state of control. CPP, IPC and yield results for the three batches shall be tabulated and compared side by side. Statistical evaluation of process variability and capability will be performed under ongoing process verification (OPV) once enough commercial batch data are available.

Cleaning validation, stability, characterization and other technical study results are reported under their own protocols and are not evaluated in this report. Release of LSN3832099/ASYM-140851 for further processing does not depend on approval of the summary report (refer to **Section 20**).

21. Batch Release and Disposition

LSN3832099/ASYM-140851 (peptide on-resin) is an in-process intermediate without a release specification. It is not distributed outside the manufacturing site; it is forward processed into Step 2 soft cleavage (ASYM-140852). The in-process controls in **Table 6** take the place of release testing for this material (refer to VP-14002101.01).

Each PPQ batch of LSN3832099/ASYM-140851 may be released for further processing when QA confirms all of the following:

IPCs: all IPCs in **Table 9** have met their acceptance criteria.

Batch documentation: QA has reviewed the executed batch record sections available at the time of release. As a minimum, this covers material charges, CPP entries and IPC results. All CPPs were within their PARs, or any excursion has been recorded as a deviation.

Deviations: all deviations for the batch have been documented and assessed for impact on quality. No critical or major deviation remains open unless QA has approved a documented impact assessment that allows processing to continue.

Equipment cleaning: if the batch was manufactured in equipment just after a cleaning validation run, QA has approved the interim cleaning validation report for that run (refer to **Section 17**).

Packaging and storage: the material has been packaged, labelled and stored as described in **Section 14**. It will be processed further within the hold time supported by the stability data in **Section 15**.

Appendix 1 Training SOP (Current Version)

Title	Procedure Number
ATR Result Investigation	QA-032 (Current version)
Batch operation record management procedure (DS)	DS-QA-001 (Current version)
Cleaning Validation (DS)	DS-GMP-043 (Current version)
Change Control	QA-006 (Current version)
Deviation investigation	QA-010(Current version)
Good record practice, good documentation practice, good review practice, and document integrity	ASM-015 (Current version)
Material homogeneity study procesure(DS)	DS-GMP-036 (Current version)
OOS Result Investigation	QA-015 (Current version)
Process Validation (DS)	DS-GMP-035 (Current version)
Product Release (DS)	DS-QA-004 (Current version)
Receipt, Release and Distribution of Raw Material for Production(DS)	DS-MM-004 (Current version)
Sampling and testing for raw material and product (DS)	DS-AC-004 (Current version)

Title	Procedure Number
Stability Testing Management Procedure (DS)	DS-AC-003 (Current version)
Sampling and Testing Procedure for Cleaning Test	AC-053 (Current version)
Supplier Control	MM-046 (Current version)
Validation and qualification of facility, utility and equipment	GMP-026 (Current version)

Appendix 2 Reference document list

Item	Title	Document and version number
1	Master Operation Record	MOR-TJ414002101.04
2	Equipment Suitability Assessment for ASYM-140021 (Step 1)	RA-140021030101.01
3	PAR report	RPT-1193831 (Lilly document number.)
4	Process Flow Document	WI-61121.(Lilly document number.)
5	Analytical Method Validation/Transfer/Verification Master Plan Protocol	PSP-140021-2026-001.01
6	Process Validation Master Plan	VP-14002101.01
7	The change of specifications for raw material amino acids in project CPo140019-03, CPo140020-03 and CPo140021-03	Change-TJ420260080.01
8	Cleaning Validation Protocol	CVP-1400210301.01
9	Stability Study Protocol of LSN3832099/ASYM-140851	PSP-140021-2026-002.01

Appendix 3 IPCs Results Summary Table

No	Sample Description	Test Item	Acceptance Criteria	CPo140021-03-01-02	CPo140021-03-01-03	CPo140021-03-01-04
1	KF after DMF washing before resin swelling	KF	NMT 500 ppm			
2	Bromophenol-blue colorimetric test in amino acid coupling (AA2)	Bromophenol blue test	Negative			
3	Bromophenol-blue colorimetric test in amino acid coupling (AA2)	Bromophenol blue test	Negative			
4	Bromophenol-blue colorimetric test in amino acid coupling (AA3)	Bromophenol blue test	Negative			

No	Sample Description	Test Item	Acceptance Criteria	CPo140021-03-01-02	CPo140021-03-01-03	CPo140021-03-01-04
5	Bromophenol-blue colorimetric test in amino acid coupling (AA3)	Bromophenol blue test	Negative			
6	Bromophenol-blue colorimetric test in amino acid coupling (AA4)	Bromophenol blue test	Negative			
7	Bromophenol-blue colorimetric test in amino acid coupling (AA4)	Bromophenol blue test	Negative			
8	Bromophenol-blue colorimetric test in amino acid coupling (AA5)	Bromophenol blue test	Negative			
9	Bromophenol-blue colorimetric test in amino acid coupling (AA5)	Bromophenol blue test	Negative			
10	Bromophenol-blue colorimetric test in amino acid coupling (AA6)	Bromophenol blue test	Negative			
11	Bromophenol-blue colorimetric test in amino acid coupling (AA6)	Bromophenol blue test	Negative			
12	Bromophenol-blue colorimetric test in amino acid coupling (AA7)	Bromophenol blue test	Negative			
13	Bromophenol-blue colorimetric test in amino acid coupling (AA7)	Bromophenol blue test	Negative			

No	Sample Description	Test Item	Acceptance Criteria	CPo140021-03-01-02	CPo140021-03-01-03	CPo140021-03-01-04
14	Bromophenol-blue colorimetric test in amino acid coupling (AA8)	Bromophenol blue test	Negative			
15	Bromophenol-blue colorimetric test in amino acid coupling (AA8)	Bromophenol blue test	Negative			
16	Bromophenol-blue colorimetric test in amino acid coupling (AA9)	Bromophenol blue test	Negative			
17	Bromophenol-blue colorimetric test in amino acid coupling (AA9)	Bromophenol blue test	Negative			
18	Bromophenol-blue colorimetric test in amino acid coupling (AA10)	Bromophenol blue test	Negative			
19	Bromophenol-blue colorimetric test in amino acid coupling (AA10)	Bromophenol blue test	Negative			
20	ASYM-140850 in post oxidation washing	ASYM-140850	≤ 0.5%			
21	LOD under drying	LOD	≤ 9.0%			

Note: Upon completion of production, Asymchem shall summarize all test results and compile them into the report.

Appendix 4 CPPs Results Summary Table

Step	Parameter	Type	Target	PAR	NOR	Unit	CPo140021-03–01-02	CPo140021-03-01-03	CPo140021-03-01-04
Step 1 SPPS	Gly10 Coupling temperature	CPP	25.0	**17.0 – 33.0**	20.0~30.0	°C			
	Post Gly10 capping DIPEA addition temperature	CPP	20.0	**15.0 – 25.0**	17.0~23.0	°C			
	Piperidine in DMF quantity (Thr9)	CPP	7.44	**6.58** – 8.39	7.22~7.66	kg/kg			
	Piperidine in DMF quantity (Asn3)	CPP	7.44	**6.77** – 8.40	7.22~7.66	kg/kg			
	DeFmoc temperature (Asn3)	CPP	25.0	**16.3** – 37.0	20.0~30.0ºC	°C			
	Total DeFmoc time (Thr6)	CPP	120	**62** – 180	110~150	min			
	Total DeFmoc time (Asn3)	CPP	120	**77** – 180	110~150	min			
	AA quantity (Cys7, Cys2)	CPP	2.00	**1.85 – 2.13**	1.94~2.06	eq			
	DMF activation total quantity (Thr9)	CPP	4.74	4.14 – **5.27**	4.60~4.88	kg/kg			
	DMF activation total quantity (Ala8, Thr6 to Asn3)	CPP	6.63	5.83 – **7.38**	6.43~6.83	kg/kg			
	DMF activation total quantity (Cys7, Cys2)	CPP	6.63	5.81 – **7.38**	6.43~6.83	kg/kg			
	DMF coupling total quantity (Thr9)	CPP	8.53	7.45 – **9.49**	8.27~8.79	kg/kg			
	DMF coupling total quantity (Ala8, Thr6 to Asn3)	CPP	7.58	6.68 – **8.43**	7.35~7.81	kg/kg			
	DMF coupling total quantity (Cys7, Cys2)	CPP	8.53	**7.71– 9.26**	8.27~8.79	kg/kg			
	Coupling temperature (Thr9, γGlu1)	CPP	25.0	**17.0 – 33.0**	20.0~30.0	°C			
	Coupling temperature (Ala8, Thr6 to Asn3)	CPP	25.0	**17.1 – 33.0**	20.0~30.0	°C			
	Coupling temperature (Cys7, Cys2)	CPP	20.0	**12.0 – 28.0**	15.0~25.0	°C			
	Oxidation temperature	CPP	-5	**-15 – 5**	-10.0~0.0	°C			

Step	Parameter	Type	Target	PAR	NOR	Unit	CPo140021-03–01-02	CPo140021-03-01-03	CPo140021-03-01-04
	Iodine solution transfer time	CPP	50	**20 – 100**	30~80	min			
	Oxidation Reaction time	CPP	0	0 – **30**	0~25	min			
	Oxidation Exposure time	CPP	50	**20** – 130	30~105	min			
	1st Post oxidation wash temperature	CPP	-3	**-7.0 – 5.0**	-5.0~0.0	℃			